PERIPHERAL HARDWARE INTEGRATION

Barcode & QR Scanner Pairing with Nexus Inventory

Technical guide for configuring 1D laser and 2D QR image scanners, programming carriage return suffixes, and optimizing high-volume retail POS checkout.

CLASSIFICATION	Hardware Integration	RELEASE VERSION	v2.4.0 (Enterprise)
AUTHOR / TEAM	Nexus Hardware Engineering	EFFECTIVE DATE	October 2026
TARGET AUDIENCE	POS Cashiers, IT Technicians, Warehouse Stock Clerks	STATUS	Official Production Ready

1. Supported Scanner Types & Technologies

Nexus ERP connects seamlessly with all standard USB HID (Human Interface Device) keyboard-wedge barcode readers. Because the scanner acts as an ultra-fast keyboard emulator, no proprietary software drivers are required.

Scanner Category	Optimal Use Case	Supported Symbologies	Connection Type
`1D Handheld Laser`	Supermarket checkout, apparel retail	EAN-13, UPC-A, Code 128, Code 39	USB Cable / 2.4GHz Wireless
`2D Image Imager`	Mobile screen reading, pharmacy QR	All 1D + QR Code, DataMatrix, PDF417	USB / Bluetooth / 2.4GHz
`Omnidirectional Desktop`	High-throughput grocery counters	Fast multi-angle 1D/2D scanning	USB / RS232
`Wireless Handheld Gun`	Warehouse stocktaking & bin audits	Long range 1D/2D barcodes (up to 50m)	Bluetooth / Base Cradle

2. Mandatory Hardware Programming: Auto-Enter Suffix

For Nexus POS to instantly add an item to the shopping cart upon scanning without requiring the cashier to press 'Enter' on the physical keyboard, the barcode scanner must be programmed to transmit a **Carriage Return (CR + LF)** suffix:

Step 1: Locate the quick-programming manual provided in your barcode scanner's physical packaging.

Step 2: Scan the configuration barcode labeled **'Enter Setup / Enter Configuration Mode'**.

Step 3: Scan the barcode labeled **'Add CR / LF Suffix'** or **'Enable Enter Key Suffix'**.

Step 4: Scan the barcode labeled **'Exit & Save Configuration'**.

Testing Auto-Enter in Notepad

To verify that the scanner is correctly programmed: Open Windows Notepad. Scan any standard barcode. If the cursor immediately jumps down to the next line automatically, your scanner is configured correctly for high-speed Nexus POS billing.

3. Nexus POS High-Speed Checkout Workflow

Once configured, cashier workflow operates with maximum throughput:

- **Single Item Scan:** Cashier scans item → product name, unit price, and stock position appear instantly; quantity increments to `1`.
- **Repeated Scan:** Scanning the same item again automatically increments quantity to `2`, `3`, etc., without modal popups.
- **Multi-Price Batch Detection:** If an item has multiple active cost/selling batches, Nexus POS displays a quick-selection modal for the cashier to choose the appropriate batch.
- **Weighing Scale Barcodes:** Nexus ERP supports embedded-price barcodes (e.g., `20XXXXX` prefixes generated by deli and butchery weighing scales), automatically parsing weight and total price.

4. Warehouse Stock Audits & Bin Relocation Scanning

In warehouse and backroom inventory operations, barcode scanners streamline stock movement:

1. **Goods Receiving (GRN):** Scan supplier outer barcodes to instantly verify against authorized Purchase Orders.
2. **Bin Relocation (ITV):** Scan the origin bin barcode, scan the product barcode, and scan the destination bin barcode for atomic relocation.
3. **Periodic Shelf Counts:** Cashiers scan shelf items sequentially; variances are calculated automatically in the Stock Bin Card.

5. Troubleshooting Scanner Recognition Issues

Issue Observed	Likely Cause	Corrective Action
`Scanner beeps but nothing types`	USB port power sleep or bad cord	Switch to rear motherboard port; disable USB power suspend.
`Characters scrambled or accented`	Windows keyboard layout mismatch	Set Windows keyboard language to English (US).
`Barcodes scan without Enter jump`	Missing CR/LF suffix programming	Scan 'Add Enter Suffix' barcode from scanner manual.
`Cannot read phone screen barcodes`	Laser scanner used instead of 2D	Upgrade cashier terminal to 2D CMOS Image Imager.